

Exact Estimation of Exponential Family Models on Massive Mixed Categorical–Continuous Data via Tilted Cell Moments

Andreas Ljungström*

Swedish Institute for Social Research, Stockholm University

July 2026

Abstract

When every covariate of a generalized linear model is categorical, the log likelihood collapses onto within-cell aggregates of the cross-classification, and estimation on the J cells reproduces the microdata maximum-likelihood estimator at a fraction of the cost. A single continuous covariate destroys this collapse: we show that computing the likelihood then requires, in effect, the entire within-cell empirical distribution of the continuous covariates, so no fixed-dimensional cell summary suffices. The contribution of this paper is that a moving summary does: for exponential dispersion families, the exact score vector and Fisher information matrix at any parameter value are linear combinations of three low-order moments of an *exponentially tilted* within-cell measure—for the Poisson family, an Esscher transform of the within-cell exposure measure. Consequently every Newton or Fisher-scoring iteration is computable from one streaming pass over the microdata at $O(Np_c^2)$ cost, with p_c the number of continuous covariates only, followed by algebra on the cells; the resulting iterates are algebraically *identical* to full-design iterations, not approximations to them. We give the streaming algorithm, its complexity and numerical properties, and exact cell-accumulated robust and cluster-robust covariance estimators. Experiments verify iterate-for-iterate agreement with dense Newton at 10^{-13} – 10^{-15} across Poisson, logistic, and gamma models while running one to two orders of magnitude faster than dense and general-purpose implementations and several times faster than a sparse-design IRLS, with exactness unaffected by severe cell imbalance or rare events. Two applications—a logistic on-time-performance model for the 336,776 flights of the public `nycflights13` data, and a register-scale synthetic epidemiology study with five million person-year episodes and 10,000 absorbed municipality–year effects—fit with inference in seconds. An online supplement develops companion theory: global convergence and sharp local rates for the nonlinear

*Swedish Institute for Social Research (SOFI), Stockholm University, SE-106 91 Stockholm, Sweden. E-mail: andreas.ljungstrom@sofi.su.se. Reproduction code for all numerical results is available at <https://github.com/andreasljungstrm/stpois> (directory `paper/stats/code`). An online supplement contains the companion theory (convergence of nonlinear fixed-effect absorption and its rates, high-dimensional asymptotics, Firth and penalized extensions) and its proofs and simulations.

alternating-projections absorption of high-dimensional fixed effects, asymptotics up to the proportional regime $J/N \rightarrow c$, and Firth, penalized, and conditional-likelihood extensions.

Keywords: generalized linear models; exponential tilting; sufficient statistics; streaming algorithms; high-dimensional fixed effects; register data.

1 Introduction

A large share of applied statistical work now takes place on data sets whose size is set not by a sampling budget but by an administrative process: full-population registers, insurance claim files, electronic health records, transportation and digital trace data. The models fitted to such data are rarely exotic—Poisson regression for event rates, logistic regression for binary outcomes, gamma regression for costs—but their covariate structure has a characteristic shape. A small number of continuous covariates of scientific interest (income, exposure, a polygenic score) appear alongside a rich categorical apparatus: demographic strata such as age band $\times$ sex $\times$ education, operational strata such as carrier $\times$ route $\times$ month, and often fixed-effect factors with thousands of levels. The cross-classification of the categorical covariates partitions the N observations into J cells, and it is the interplay between N , J , and the continuous covariates that determines whether maximum-likelihood estimation is an interactive computation or an overnight batch job.

Two classical observations frame the problem. First, when *every* covariate is categorical, the exponential-family log likelihood depends on the data only through within-cell aggregates; estimation on the J collapsed cells reproduces the microdata maximum-likelihood estimator (MLE) exactly, and the per-iteration cost drops from $O(Nq^2)$ to $O(Jq^2)$ for q parameters. This is the standard device behind register-based rate modeling in epidemiology and demography (Holford, 1980; Laird and Olivier, 1981; Clayton and Hills, 1993). Second, the collapse is fragile: a single continuous covariate destroys it, because the cumulant function of the family is strictly convex, so cell sums of nonlinear transforms of the linear predictor cannot be recovered from cell means. Analysts then face an unattractive choice between discretizing the continuous covariates—introducing bias that is hard to quantify—and abandoning the cell structure entirely, paying the full $O(Nq^2)$ price per iteration even though most of the q parameters are attached to covariates that are constant within cells. General-purpose GLM software pays this price by default; software for high-dimensional fixed effects (Guimarães and Portugal, 2010; Gaure, 2013; Correia et al., 2020; Bergé, 2018; Stammann, 2018) removes *dummy-coded factors* from the design by demeaning, but continuous-by-cell interplay of the kind above is not a demeaning problem, and the cell structure of the categorical part goes unexploited.

This paper develops a resolution that is exact rather than approximate. The starting point is a precise statement of what breaks (Proposition 1): with continuous covariates, any statistic from which the likelihood *function* can be evaluated must determine the entire within-cell empirical distribution of the continuous covariates—the within-cell Laplace transform appears in the likelihood, and Laplace transforms determine measures. No fixed cell summary can therefore support the full likelihood surface. The contribution is that estimation does not need the full surface; it needs score and curvature *at the current iterate*, and these have an exact finite representation. Theorem 1 shows that for exponential dispersion families with canonical

link, the score vector and Fisher information at any parameter value are linear combinations of the zeroth-, first-, and second-order moments of an *exponentially tilted probability measure* on each cell—for the Poisson family, an Esscher transform of the within-cell empirical exposure measure, connecting the construction to classical tilting ideas in saddlepoint approximation and empirical likelihood (Esscher, 1932; Daniels, 1954; Efron, 1981; Jensen, 1995; Kitamura and Stutzer, 1997). The tilt is the right *dynamic* object: it changes with the iterate, which is exactly why a fixed collapse fails and a per-iteration one succeeds. Corollary 1 converts the identity into complexity: each Newton iteration costs one streaming $O(N(p^2 + p))$ pass over the microdata— p the number of continuous columns only—plus $O(J(k^2 + kp))$ cell algebra, against $O(N(p + k)^2)$ for the full design, a saving that grows quadratically in the richness k of the categorical structure. Because the identity is exact, the accelerated iterates are *identical* to full-design Newton—not close, identical—which Section 5.1 verifies at the 10^{-13} – 10^{-15} level, iterate by iterate, across Poisson, logistic, and gamma models. Fisher scoring under non-canonical links has the same structure (Proposition 2); for the gamma–log family the tilt weights are identically one and the moments are constant across iterations.

The second half of the paper is deliberately computational. Section 4 gives the streaming algorithm and its implementation envelope: numerical safeguards (step-halving, log-domain evaluation, rank-deficiency fallbacks), memory layout and parallelization of the bandwidth-bound accumulation pass, exact cell-accumulated robust and cluster-robust covariance estimators, and the combination with absorbed fixed-effect factors, for which the companion theory—global convergence of the nonlinear alternating-projections cycle, its local rates for two and for three or more factors, and high-dimensional asymptotics up to one fixed effect per two observations—is developed in the online supplement. Section 5 contains the experiments: iterate-level exactness and per-iteration cost against dense Newton (Section 5.1); total-time benchmarks against **statsmodels** (the standard general-purpose Python GLM) and against a sparse-design IRLS, the natural “just use sparse matrices” alternative (Section 5.2); and sensitivity to severe cell imbalance and rare events (Section 5.3). Section 6 applies the method twice: to the public **nycflights13** data (a logistic on-time-performance model with a carrier–origin–destination–month cell structure, verified coefficient-for-coefficient against **statsmodels** at a tenth of its cost), and to a fully reproducible register-scale synthetic study (five million person-year episodes, 10,000 absorbed municipality–year effects, cluster-robust inference, seconds on one core). Section 7 summarizes and points to the supplement’s theory. Proofs of the main-paper results are in the appendix; every number regenerates from short scripts included with the manuscript.

Relative to the literature, the closest computational ideas are the collapsed sufficient statistics of classical log-linear rate modeling (Holford, 1980; Clayton and Hills, 1993), which the tilted moments generalize from “all categorical” to “categorical plus continuous”; and the alternating-projections HD FE stack (Guimarães and Portugal, 2010; Gaure, 2013; Correia, 2016; Correia et al., 2020; Bergé, 2018), which removes dummy factors but streams the full non-absorbed

design through every IRLS step—the two devices are complementary, and Section 4.5 combines them. The framework is implemented in the Stata command `stpois` for the Poisson event-history case and in an open-source Python reference implementation used for all experiments here.

2 Exponential family models and the failure of ordinary collapse

2.1 Model and notation

Observations $i = 1, \dots, N$ carry a response y_i , a vector $\mathbf{x}_i \in \mathbb{R}^p$ of *individual-level* covariates (those involving at least one continuous variable, including continuous–categorical interaction columns), a vector $\mathbf{w}_{j(i)} \in \mathbb{R}^k$ of *cell-level* covariates, and a known offset o_i . The map $j : \{1, \dots, N\} \rightarrow \{1, \dots, J\}$ assigns each observation to the cell of the cross-classification of the categorical covariates; $\mathbf{w}_j$ is constant within cell j by construction. Conditional on the covariates, responses are independent draws from an exponential dispersion family

$$f(y_i \mid \theta_i, \phi) = \exp \left\{ \frac{y_i \theta_i - b(\theta_i)}{\phi} + c(y_i, \phi) \right\}, \quad (1)$$

with $b : \Theta \rightarrow \mathbb{R}$ strictly convex and three times continuously differentiable on the open canonical parameter space $\Theta \subseteq \mathbb{R}$, and dispersion $\phi > 0$ (McCullagh and Nelder, 1989). The mean and variance are $\mu_i = b'(\theta_i)$ and $\text{Var}(y_i) = \phi b''(\theta_i)$. We model the linear predictor

$$\eta_i = \boldsymbol{\gamma}' \mathbf{x}_i + \boldsymbol{\delta}' \mathbf{w}_{j(i)} + o_i, \quad \boldsymbol{\theta} = (\boldsymbol{\gamma}', \boldsymbol{\delta}')' \in \mathbb{R}^q, \quad q = p + k, \quad (2)$$

and, until Section 3.2, take the link to be canonical, $\theta_i = \eta_i$. The Poisson event-history model that motivates much of the applied work is the special case $b(\theta) = e^\theta$, $\phi = 1$, $y_i \in \{0, 1, 2, \dots\}$ an event count, and $o_i = \log t_i$ the log of the time at risk; the piecewise-exponential equivalence of Holford (1980); Laird and Olivier (1981); Whitehead (1980) makes this the canonical likelihood for register-based survival analysis. Up to terms not involving $\boldsymbol{\theta}$, the log likelihood is

$$\ell(\boldsymbol{\theta}) = \frac{1}{\phi} \sum_{i=1}^N \{y_i \eta_i - b(\eta_i)\}. \quad (3)$$

Because (3) is a sum of concave functions of the linear predictor, ℓ is concave on $\mathbb{R}^q$, strictly so on the quotient of $\mathbb{R}^q$ by the null space of the stacked design $\mathbf{a}_i = (\mathbf{x}_i', \mathbf{w}_{j(i)}')'$. We write $\mathbf{U}(\boldsymbol{\theta}) = \nabla \ell$ for the score, $\mathbf{H}(\boldsymbol{\theta}) = -\nabla^2 \ell$ for the negative Hessian, and note that for canonical links $\mathbf{H}$ equals the Fisher information $\mathbf{F}(\boldsymbol{\theta}) = \mathbb{E}[-\nabla^2 \ell]$, so Newton–Raphson and Fisher scoring coincide.

2.2 Cell collapse and its failure

Suppose first that $p = 0$: every covariate is categorical, so $\eta_i = \boldsymbol{\delta}'\mathbf{w}_j + o_i$ varies over i within a cell only through the offset. For the Poisson family with $o_i = \log t_i$, (3) becomes $\ell(\boldsymbol{\delta}) = \sum_j \{D_j \boldsymbol{\delta}'\mathbf{w}_j - T_j e^{\boldsymbol{\delta}'\mathbf{w}_j}\} + \text{const}$, with $D_j = \sum_{i \in j} y_i$ and $T_j = \sum_{i \in j} t_i$: the collapsed cell data (D_j, T_j) carry all the information, and estimation on the J cells is exact. The following proposition records why no analogous finite collapse survives the introduction of a continuous covariate, and in what precise sense. Define the within-cell *exposure measure* $\nu_j = \sum_{i \in j} e^{o_i} \delta_{\mathbf{x}_i}$ on $\mathbb{R}^p$ (for Poisson, $\nu_j = \sum_{i \in j} t_i \delta_{\mathbf{x}_i}$), a finite discrete measure recording the joint configuration of exposures and continuous covariates in cell j .

Proposition 1 (Collapse and its failure). *Consider the Poisson family with offsets. (i) If $\mathbf{x}_i = \bar{\mathbf{x}}_j$ for all $i \in j$ and all j , the log likelihood (3) depends on the data only through $\{(D_j, T_j, \bar{\mathbf{x}}_j, \sum_{i \in j} y_i \mathbf{x}_i)\}_{j \leq J}$, and the MLE on the collapsed cells equals the microdata MLE. (ii) In general, $\ell(\boldsymbol{\gamma}, \boldsymbol{\delta})$ depends on the data through $\{S_j, D_j, \mathcal{L}_{\nu_j}\}_{j \leq J}$, where $S_j = \sum_{i \in j} y_i \mathbf{x}_i$ and $\mathcal{L}_{\nu_j}(\boldsymbol{\gamma}) = \int e^{\boldsymbol{\gamma}'\mathbf{x}} \nu_j(d\mathbf{x}) = \sum_{i \in j} t_i e^{\boldsymbol{\gamma}'\mathbf{x}_i}$ is the Laplace transform of ν_j ; and $\mathcal{L}_{\nu_j}$ determines ν_j . Consequently any statistic from which the likelihood function can be computed for all $\boldsymbol{\gamma}$ in an open set must determine the entire within-cell exposure measure, whose support size—hence the dimension of any such statistic—grows with the number of distinct covariate values in the cell. (iii) By strict convexity of the exponential (Jensen), $\mathcal{L}_{\nu_j}(\boldsymbol{\gamma}) \geq T_j e^{\boldsymbol{\gamma}'\bar{\mathbf{x}}_j}$ with equality for some $\boldsymbol{\gamma} \neq \mathbf{0}$ direction only if $\mathbf{x}$ is ν_j -a.s. constant along that direction: replacing microdata by cell means strictly understates the within-cell mass of the cumulant term and is not a reparametrization of the same likelihood.*

Part (ii) is the formal statement of the practical dictum that “continuous covariates destroy collapsibility”: the fixed-dimensional sufficient statistic $\sum_i y_i \mathbf{a}_i$ for the canonical parameter still exists, but *computing the likelihood*—and hence the MLE—requires functionals of the data whose dimension is of order N , not J . The next subsection shows that the required functionals nevertheless have a special structure: for each trial value of $\boldsymbol{\gamma}$ they are three low-order moments of a tilted measure, computable in one linear pass.

3 Tilted cell moments

3.1 Exponentially tilted cell moments

Definition 1 (Tilted within-cell measures). *Fix $\boldsymbol{\theta} = (\boldsymbol{\gamma}, \boldsymbol{\delta})$ and let $\omega_i = b''(\eta_i) > 0$ be the information weight of observation i . The information-tilted measure on cell j is the probability measure*

$$P_{j,\boldsymbol{\theta}}(\{i\}) = \frac{\omega_i}{\sum_{i' \in j} \omega_{i'}}, \quad i \in j.$$

For the Poisson family, $\omega_i = \mu_i = t_i e^{\delta' \mathbf{w}_j + \gamma' \mathbf{x}_i}$, the factor $e^{\delta' \mathbf{w}_j}$ cancels, and $P_{j,\boldsymbol{\theta}}$ reduces to the Esscher transform (exponential tilt) of the normalized exposure measure ν_j/T_j :

$$P_{j,\gamma}(\mathrm{d}\mathbf{x}) = \frac{e^{\gamma' \mathbf{x}} \nu_j(\mathrm{d}\mathbf{x})}{\mathcal{L}_{\nu_j}(\gamma)}, \quad (4)$$

free of $\boldsymbol{\delta}$. Its moments are derivatives of the within-cell cumulant generating function: with $K_j(\gamma) = \log \mathcal{L}_{\nu_j}(\gamma)$, $\mathbb{E}_{P_{j,\gamma}}[\mathbf{x}] = \nabla K_j(\gamma)$ and $\text{Var}_{P_{j,\gamma}}(\mathbf{x}) = \nabla^2 K_j(\gamma)$.

The tilt (4) is precisely the change of measure that underlies saddlepoint approximation (Daniels, 1954; Jensen, 1995) and exponential-tilting estimators in the empirical-likelihood family (Efron, 1981; Kitamura and Stutzer, 1997; Schennach, 2007); here it appears not as an approximation device but as an exact algebraic identity for the score and information, as follows. Write, for each cell j ,

$$M_j^{(0)} = \sum_{i \in j} \omega_i, \quad \mathbf{m}_j = \sum_{i \in j} \omega_i \mathbf{x}_i, \quad \mathbf{Q} = \sum_{i=1}^N \omega_i \mathbf{x}_i \mathbf{x}_i', \quad (5)$$

and let $B_j = \sum_{i \in j} b'(\eta_i)$ and $R_j = \sum_{i \in j} (y_i - b'(\eta_i))$ denote the cell sums of means and of raw residuals.

Theorem 1 (Exact tilted-moment representation). *For the model (1)–(2) with canonical link, at every $\boldsymbol{\theta} \in \mathbb{R}^q$:*

(i) *The score is*

$$\phi \mathbf{U}(\boldsymbol{\theta}) = \begin{pmatrix} \sum_i y_i \mathbf{x}_i - \sum_j \mathbf{m}_j^{b'} \\ \sum_j \mathbf{w}_j R_j \end{pmatrix}, \quad \mathbf{m}_j^{b'} = \sum_{i \in j} b'(\eta_i) \mathbf{x}_i = B_j \mathbb{E}_{P_{j,\boldsymbol{\theta}}^{b'}}[\mathbf{x}],$$

where $P_{j,\boldsymbol{\theta}}^{b'}$ is the tilt of Definition 1 with b' in place of b'' .

(ii) *The information matrix is*

$$\phi \mathbf{F}(\boldsymbol{\theta}) = \begin{pmatrix} \mathbf{Q} & \sum_j \mathbf{m}_j \mathbf{w}_j' \\ \sum_j \mathbf{w}_j \mathbf{m}_j' & \sum_j M_j^{(0)} \mathbf{w}_j \mathbf{w}_j' \end{pmatrix}, \quad \mathbf{m}_j = M_j^{(0)} \mathbb{E}_{P_{j,\boldsymbol{\theta}}}[\mathbf{x}], \quad \mathbf{Q} = \sum_j M_j^{(0)} \mathbb{E}_{P_{j,\boldsymbol{\theta}}}[\mathbf{x} \mathbf{x}'].$$

(iii) *For the Poisson family, $b' = b''$, the two tilts coincide with the $\boldsymbol{\delta}$ -free Esscher transform (4), and $B_j = M_j^{(0)} = e^{\delta' \mathbf{w}_j} \mathcal{L}_{\nu_j}(\gamma)$, $\mathbf{m}_j = M_j^{(0)} \nabla K_j(\gamma)$, and the γ -block of $\mathbf{F}$ is $\sum_j M_j^{(0)} \{\nabla^2 K_j(\gamma) + \nabla K_j \nabla K_j'\}$.*

(iv) *The Newton step $\boldsymbol{\theta}^+ = \boldsymbol{\theta} + \mathbf{F}(\boldsymbol{\theta})^{-1} \mathbf{U}(\boldsymbol{\theta})$ computed from the quantities in (i)–(ii) is identical to the Newton step computed from the full $N \times q$ design.*

Claim (iv) is trivial once (i)–(ii) are established—the two computations produce the same matrices—but it is the operative point: the decomposition involves *no linearization, discretization, or approximation of any kind*, so the accelerated algorithm is not a fast approximate estimator but a fast exact organization of the arithmetic of the MLE. Section 5.1 exhibits agreement of the full iterate *sequences* at the 10^{-13} level, which is the accumulation of floating-point reordering only.

Statistically, the representation has content beyond bookkeeping. The δ -score in (i) compares each cell’s observed event mass with its expected mass under the current tilt; the cross-information in (ii) is the tilted covariance between cell-level and individual-level design; and the γ -block in (iii) decomposes into a within-cell tilted covariance $\nabla^2 K_j$ plus a between-cell term. In particular the identification of γ rests exactly on $\sum_j M_j^{(0)} \nabla^2 K_j(\gamma) \succ \mathbf{0}$: continuous covariates are identified through their *within-cell, tilt-weighted* variation—a fact that the block structure displays, and that drives the Schur-complement profile Hessians used throughout the companion theory in the online supplement. (When a continuous covariate is constant within cells—a route-level distance, say—it remains identified through between-cell variation as long as the cell-level design $\mathbf{w}_j$ is not saturated; Section 6.1 exercises this case.)

Corollary 1 (Per-iteration complexity). *One Newton iteration computed via Theorem 1 costs*

$$\underbrace{O\left(N(p^2 + p + 1)\right)}_{\text{microdata pass}} + \underbrace{O\left(J(k^2 + kp)\right)}_{\text{cell assembly}} + \underbrace{O\left((p + k)^3\right)}_{\text{solve}}$$

floating-point operations and one $O(N)$ memory sweep, versus $O(N(p + k)^2) + O((p + k)^3)$ for the full-design iteration. When $N \gg J(p + k)$ the leading-order saving is the factor $(p + k)^2/(p^2 + p + 1)$, which grows quadratically in the richness k of the categorical structure; when additionally $p = 0$ the microdata pass degenerates to the classical cell collapse and the entire iteration runs at $O(Jk^2)$.

Two remarks on the constant factors. First, the microdata pass is a fused stream: one evaluation of $b'(\eta_i)$, $b''(\eta_i)$ per observation, $p + 1$ multiply–accumulate reductions into cell bins, and a rank-one update of the $p \times p$ Gram matrix; on modern hardware it is memory-bandwidth-bound, so measured speedups (Section 5.1) are large but smaller than the FLOP ratio. Second, the accumulations are plain segmented sums—**bincount** operations—so they parallelize and stream trivially; no $N \times q$ design matrix is ever formed, which converts a memory-infeasible problem (Section 6.2) into an in-cache one.

3.2 Non-canonical links

For a non-canonical, three-times-differentiable, strictly monotone link g with $\mu_i = g^{-1}(\eta_i)$, define the usual Fisher weight and working residual

$$\omega_i = \frac{1}{V(\mu_i) g'(\mu_i)^2}, \quad r_i = (y_i - \mu_i) g'(\mu_i) \omega_i,$$

with V the variance function. Fisher scoring replaces $(b'(\eta_i), b''(\eta_i))$ by (μ_i, ω_i) and the raw residual by r_i .

Proposition 2 (Tilted Fisher scoring). *With ω_i and r_i as above, the expected-information scoring step admits the representation of Theorem 1 with $M_j^{(0)}$, $\mathbf{m}_j$, $\mathbf{Q}$ built from ω_i and with $R_j = \sum_{i \in j} r_i$; the fixed points of the resulting iteration are exactly the roots of the ML score equations, and Corollary 1 applies verbatim. For the gamma family with log link, $V(\mu) = \mu^2$ and $g'(\mu) = 1/\mu$ give $\omega_i \equiv 1$: the tilted moments reduce to untilted cell moments of the covariates, fixed across iterations, so after the first pass each subsequent iteration costs $O(Np) + O(Jk^2)$ only.*

3.3 Families in scope

Three instances cover most applied use and are carried through the experiments. *Poisson with log link* (canonical): the tilt is the δ -free Esscher transform (4), and the event-history offset $o_i = \log t_i$ makes this the estimator of piecewise-constant hazard models on episode-split survival data (Holford, 1980; Whitehead, 1980). *Logistic* (canonical): the tilt weights $\omega_i = \mu_i(1 - \mu_i)$ concentrate the within-cell measure on observations near the decision boundary and depend on δ , so the moments are re-accumulated each iteration—which the single pass does anyway. *Gamma with log link* (non-canonical, Proposition 2): the Fisher weights are identically one, the moments are constant across iterations, and after the first pass each iteration costs $O(Np) + O(Jk^2)$. We deliberately stop the catalogue here; the point is the structure, not an enumeration.

4 Algorithms and implementation

4.1 The streaming pass

Algorithm 1 states the method operationally. Its preprocessing is one factorization of the categorical covariates into cell identifiers $j(i)$ and one construction of the $J \times k$ cell-level design $\mathbf{W}$; both are $O(N)$ given a sort or hash of the categorical columns, and both are shared across models on the same cells.

By Theorem 1(iv) the iterate sequence is identical to full-design Newton from the same start; every design decision below concerns constants, not correctness.

Algorithm 1 Tilted-moment Newton / Fisher scoring

- 1: Precompute cell ids $j(i)$, cell design $\mathbf{W} \in \mathbb{R}^{J \times k}$, continuous columns $\mathbf{X} \in \mathbb{R}^{N \times p}$, offsets o_i ; initialize $\boldsymbol{\theta}_0 = (\gamma_0, \boldsymbol{\delta}_0)$.
 - 2: **repeat**
 - 3: *Stream over* $i = 1, \dots, N$: compute $\eta_i = \boldsymbol{\gamma}'\mathbf{x}_i + (\mathbf{W}\boldsymbol{\delta})_{j(i)} + o_i$, the weight ω_i , and the residual r_i ; accumulate $M_{j(i)}^{(0)} += \omega_i$; $\mathbf{m}_{j(i)} += \omega_i\mathbf{x}_i$; $R_{j(i)} += r_i$; $\mathbf{U}_\gamma += r_i\mathbf{x}_i$; $\mathbf{Q} += \omega_i\mathbf{x}_i\mathbf{x}_i'$.
 - 4: *Cell assembly*: $\mathbf{U}_\delta = \mathbf{W}'\mathbf{R}$; $\mathbf{F}_{\gamma\delta} = \mathbf{m}'\mathbf{W}$; $\mathbf{F}_{\delta\delta} = \mathbf{W}'\text{diag}(M^{(0)})\mathbf{W}$; $\mathbf{F}_{\gamma\gamma} = \mathbf{Q}$.
 - 5: Solve the $(p+k)$ -dimensional system $\mathbf{F}\Delta = \mathbf{U}$; update $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \lambda\Delta$ with backtracking $\lambda \in \{1, \frac{1}{2}, \dots\}$ on the log likelihood.
 - 6: **until** $\max_\ell |\Delta\theta_\ell|/(1 + |\theta_\ell|) < \tau$
-

4.2 Numerical safeguards

Three matter in practice. (i) *Monotonicity*: the backtracking line search on the log likelihood guarantees ascent and terminates, since the Newton direction of a positive-definite system is an ascent direction; in well-conditioned problems the full step is accepted after the first iterations and convergence is quadratic. (ii) *Log-domain evaluation*: likelihood and mean evaluations use `log1p` / `logaddexp`-style kernels so that extreme linear predictors (produced, e.g., en route to a separated fit) yield $-\infty$ log likelihoods rather than overflow. (iii) *Rank deficiency*: if the assembled $\mathbf{F}$ is singular—collinear cell-level columns, or vanishing weights under quasi-separation—the solve falls back to a minimum-norm (pseudoinverse) direction and the iteration runs to its cap with the non-convergence flagged, never silently; separation is a property of the data, not of the algorithm, and Section S3 of the supplement develops the Firth-adjusted variant that restores a finite maximizer inside the same pass.

4.3 Memory layout and parallelism

The gap between measured and FLOP ratios merits a constructive comment, since closing it is an implementation exercise with a known playbook. The tilted pass streams $O(Np)$ numbers and performs $O(Np^2)$ flops, so for small p its arithmetic intensity is low and the memory bus, not the ALU, sets the speed; the dense baseline, by contrast, is a BLAS-3 Gram update running near peak. Three standard remedies raise the tilted pass toward bandwidth saturation and beyond: (i) *cache locality*: storing the microdata sorted by cell converts the binned accumulations into contiguous segmented sums, so each cell’s partial moments stay in registers/L1 and the pass touches each cache line once (this is how the Mata implementation in `stpois` organizes the sweep, following [Correia, 2016](#)); (ii) *fusion*: the weight evaluation, residual, binned sums, and Gram update share one loop, so the data are read once per iteration rather than once per moment; and (iii) *parallelism*: cells are independent, so the pass shards trivially across threads with per-thread partial moments and a Jk -sized reduction, with no

atomics on the hot path—the same structure GPU segmented-reduction primitives expect. The single-threaded NumPy reference implementation uses none of (i)–(iii) beyond what `bincount` provides, so the reported $12\times$ – $117\times$ should be read as a floor; the FLOP ratio governs how the comparison scales with (p, k) , and the bandwidth analysis governs the attainable constant.

4.4 Robust and clustered variance estimation

All variance estimators used in practice share the cell structure. The observed/expected information is $\mathbf{F}$ itself (Theorem 1). The HC-type sandwich needs $\sum_i \hat{u}_i^2 \mathbf{a}_i \mathbf{a}_i'$ with $\hat{u}_i = y_i - \hat{\mu}_i$; its blocks accumulate cellwise exactly like (5) with ω_i replaced by $\hat{u}_i^2$. The cluster-robust estimator sums scores within clusters before the outer product; when clusters are unions of cells (the common design: cluster on region, cells within region) the accumulation is again one pass. These identities are what make *inference*, not just point estimation, run at cell-collapsed cost; they are implemented in the reference code and validated against dense computation in the test suite.

4.5 Absorbed fixed-effect factors

Applications often add factors whose level counts preclude even cell-level treatment—firms, hospitals, municipality–year interactions. The framework composes with the standard absorption device: augment the linear predictor with G absorbed factors $\alpha_{g(i)}^{(g)}$, concentrate them out by cycling closed-form profile updates (for Poisson, $\alpha_a^{(g)} = \log\{\sum_{i:g(i)=a} y_i / \sum_{i:g(i)=a} \exp(\eta_i - \alpha_a^{(g)})\}$), the multiplicative update of [Bergé, 2018](#); [Correia et al., 2020](#)), and take each tilted-moment Newton step on (γ, δ) at the profiled effects, with the inner cycle re-solved at every trial point of the line search (Algorithm S1 of the supplement). Each inner sweep is $O(N)$; with a single absorbed factor the inner solve is exact in one sweep.

The theory of this composition is the subject of the online supplement, and we summarize what is proved there. Theorem S1 establishes *global convergence* of the nonlinear cycle to the MLE whenever the MLE exists, for canonical and more generally log-concave links—a guarantee previously absent from the applied HDFE–GLM literature. Theorems S2–S3 give the *local rate*: for two absorbed factors the per-sweep contraction equals $\cos^2$ of the Friedrichs angle between the weighted factor subspaces—equivalently the squared second singular value of a normalized weighted cross-tabulation—and for $G \geq 3$ factors the exact rate is the spectral radius of a level-space sweep operator determined by the pairwise cross-tabulations alone, computable by power iteration before touching the microdata, with an angle-based product certificate; observed contraction factors match these predictions to three decimals. Theorems S4–S6 give asymptotics for the estimator (which is the MLE, so classical theory applies at fixed dimension) up to the proportional regime of one Poisson fixed effect per bounded-size cell, $J/N \rightarrow c \in (0, 1)$, where profiling coincides with conditioning and inference is unbiased with valid cell-clustered

Table 1: Study 1: tilted-moment versus dense Newton. “Iterate dev.” is the maximum absolute difference between the two iterate sequences over all iterations and coordinates; times are medians per Newton iteration. Selected configurations; the full grid is in the supplement.

Family	N	J	p	Iterate dev.	Final dev.	t_{dense} (s)	t_{tilted} (s)
Poisson	2×10^5	50	2	5.1×10^{-14}	2.4×10^{-15}	0.119	0.010
Poisson	2×10^5	200	2	5.1×10^{-14}	3.7×10^{-15}	0.503	0.010
Poisson	2×10^5	500	2	6.0×10^{-13}	3.0×10^{-14}	1.481	0.017
Poisson	10^6	200	2	7.4×10^{-14}	1.1×10^{-14}	2.490	0.062
Poisson	2×10^5	200	5	1.1×10^{-13}	1.9×10^{-15}	0.520	0.016
Logistic	2×10^5	500	2	2.2×10^{-14}	4.0×10^{-15}	1.738	0.015
Logistic	10^6	200	2	8.2×10^{-15}	2.2×10^{-16}	2.918	0.082
Gamma	2×10^5	500	2	1.3×10^{-13}	5.2×10^{-15}	1.443	0.013
Gamma	10^6	200	2	2.5×10^{-13}	1.3×10^{-15}	2.662	0.058

standard errors. Separated absorbed groups (no events) are detected and dropped a priori, exactly as in [Correia et al. \(2020\)](#); Section 6.2 exercises the full composition at scale.

5 Numerical experiments

All experiments run on a single laptop-class core (Python/NumPy reference implementation; scripts `sim1*` in the code supplement) and complete in minutes; every number below regenerates from a fixed seed. The studies validating the supplement’s absorption, asymptotic, and separation theory (convergence rates for two and three factors, the proportional regime, Firth under quasi-separation, the logit bias and its conditional repair) are reported in Section S4 of the online supplement.

5.1 Exactness and per-iteration cost

Study 1 fits Poisson, logistic, and gamma models with $J \in \{50, 200, 500\}$ cells, $p \in \{2, 5\}$ continuous covariates, and N up to 10^6 , tracing the full iterate sequence of tilted-moment Newton and of textbook dense Newton from the same starting point. Table 1 reports the maximum deviation between the two iterate sequences across all iterations, the final-estimate deviation, and the per-iteration wall-clock times.

Three observations. First, the iterate sequences agree to 10^{-13} – 10^{-15} throughout—floating-point reordering noise—confirming Theorem 1(iv): this is the same algorithm, not an approximation. Second, measured per-iteration speedups range from $12\times$ ($J = 50$) to $117\times$ ($J = 500$), growing with k as Corollary 1 predicts, though below the raw FLOP ratios ($430\times$ – $4700\times$) because the dense baseline runs BLAS-3 while the tilted pass is memory-bound—the FLOP ratio is the right guide to scaling, the measured ratio to wall-clock budgeting. Third, the dense baseline at $N = 10^6$, $J = 200$ already needs a 1.6 GB design matrix where the tilted pass needs

Table 2: Study 1b: total wall-clock time to convergence and maximum absolute coefficient deviation from the tilted fit; $N = 200,000$, $J = 200$, $p = 2$, tolerance 10^{-9} .

Family	Time (s)				Max. deviation	
	Tilted	Dense	<code>statsmodels</code>	Sparse	<code>statsmodels</code>	Sparse
Poisson	0.11	3.85	36.8	0.24	7×10^{-15}	2×10^{-15}
Logistic	0.07	1.74	24.4	0.15	3×10^{-15}	7×10^{-15}

the p continuous columns only; at the dimensions of Section 6.2 the dense fit is not slow but impossible.

5.2 Comparison with general-purpose and sparse implementations

Study 1b benchmarks total time to convergence (identical tolerance 10^{-9} , comparable starts) for the same model ($N = 200,000$, $J = 200$ cells, $p = 2$) against two external baselines: the `statsmodels` GLM implementation—IRLS on the dense design, the standard general-purpose route in Python—and a sparse-design IRLS in which the design (continuous columns plus cell dummies) is held in CSR form and the normal equations are formed and solved sparsely, i.e., the natural “the dummies are sparse, so use a sparse solver” alternative that a careful practitioner would code.

Three lessons from Table 2. First, all four implementations agree to 10^{-15} : everything here computes the same MLE, and the comparison is purely about arithmetic organization. Second, against the implementations practitioners actually reach for, the gains are larger than against our own dense baseline: two orders of magnitude over `statsmodels` (whose IRLS re-forms and factorizes the dense system each step). Third—and most informative about *where* the structure pays—the sparse-design IRLS is a respectable competitor, as it should be: cell dummies are the textbook sparse design. The tilted pass is still 2–3 $\times$ faster here because it never materializes a design at all (the sparse route stores and re-multiplies $N(p + 1)$ nonzeros and re-forms an $O((p + k)^2)$ -sparsity Gram matrix each iteration), and the gap widens with k and with interactions of categorical variables, which inflate the sparse Gram fill-in but leave the tilted cell algebra unchanged. The fair summary is not that sparse solvers fail—they do well—but that the tilted representation is what a sparse solver would discover if it also knew that the dummy block is a partition: the cell structure is sparser than sparsity.

5.3 Sensitivity: imbalance and rare events

Study 1c re-runs the exactness comparison under stress: cell sizes drawn from a Zipf(1.6) law (a few giant cells, a long tail of tiny ones) and event rates pushed to $\sim 0.2\%$ (Poisson means ~ 0.002 ; logistic prevalence $\sim 0.5\%$), with event-free cells dropped as the MLE requires. Across all eight family $\times$ scenario combinations the maximum deviation from dense Newton remains

at 10^{-15} – 10^{-14} , convergence takes at most 12 iterations (rare-event fits need a few more steps, as expected from their flatter likelihoods), and the streaming pass time is unchanged—the accumulation is a data-independent $O(N)$ sweep, so imbalance shifts work between cells without changing its total. Exactness is an algebraic identity and is expected to survive stress; the value of the study is confirming that the *implementation* (binned accumulation, line search, drop rules) inherits the robustness (script `sim1b_baselines.py`).

6 Applications

6.1 On-time performance of NYC flights

The first application is to public data, chosen so that every number can be reproduced from a fresh download: the `nycflights13` data set (Wickham, 2021)—all 336,776 flights departing New York City’s three airports in 2013, from the U.S. Bureau of Transportation Statistics. We model the probability of a late arrival (arrival delay ≥ 15 minutes, the FAA on-time criterion; 24.5% of the 327,346 flights with an observed delay) by logistic regression with a cell structure of carrier $\times$ origin $\times$ destination $\times$ month— $J = 3,863$ observed combinations, entering through a main-effects cell-level design with $k = 131$ parameters (16 carriers, 3 origins, 103 destinations, 12 months)—and $p = 2$ continuous covariates: scheduled departure time (in hours, centered) and log distance (standardized). One destination (LEX, a single on-time flight) is quasi-separated—its dummy’s MLE is $-\infty$ —and is dropped by the level-wise separation rule, after which the MLE exists.

The tilted logistic fit converges in 7 Newton iterations, 1.7 seconds including robust standard errors. The diurnal effect is the substantive headline: the odds of a late arrival multiply by 1.11 per hour of scheduled departure time (robust SE on the log scale 0.001)—delays cascade through the day, a finding familiar from the air-transportation literature, here estimated jointly with a full operational cell structure. Log distance illustrates the boundary case flagged in Section 3: distance is constant within origin–destination routes, hence within cells, so after destination main effects it is identified only through cross-route contrasts and is correspondingly imprecise (+0.65, robust SE 0.28)—the estimator handles cell-constant continuous covariates correctly and the standard error says exactly how little such a covariate adds.

As external validation, the identical model was fitted with `statsmodels` GLM on the dense $327,345 \times 133$ design (0.35 GB): maximum absolute coefficient deviation 2.3×10^{-13} , at 16.6 seconds versus 1.7—the same estimator, ten times cheaper, with no design matrix in memory (script `app_flights.py`; the data file is fetched from the public repository cited in the script).

6.2 A register-scale stress test

The second application is a stress test at the scale that motivates the framework: spatial epidemiology on an emulated population register. Because genuine Nordic register microdata cannot leave certified secure environments, the study uses a fully reproducible synthetic register, calibrated to published all-cause mortality gradients (Gompertz age slope, female advantage of ~ 0.45 on the log scale, educational gradients of 0.1–0.4, a secular decline of 1.2% per year); the generator is 40 lines of the supplement and every number below regenerates end to end in under a minute. The scientific template is a standard one—does a within-municipality environmental exposure predict mortality after fine demographic and spatial-temporal control?—and the computational shape combines every element of Sections 2–4.

The data comprise $N = 5,000,000$ person-year episodes (35,526 deaths; crude rate 9.5 per 1,000 person-years) with: municipality $\times$ year fixed effects ($500 \times 20 = 10,000$ groups) absorbed by the profile Gauss–Seidel scheme of Section 4.5 (Algorithm S1); a cell-level design of education (4) $\times$ sex (2) $\times$ age band (12), $J = 96$ cells, $k = 16$ parameters; and $p = 2$ continuous covariates—standardized log income and a standardized pollution-exposure index correlated with the municipality effects (making the fixed effects non-ignorable for it). Standard errors are clustered on the 500 municipalities.

The fit: 322 zero-event municipality–year groups (159,464 episodes) are dropped by the separation rule of Section 4.5; the remaining model—9,678 absorbed effects, 16 cell-level and 2 individual-level coefficients—converges in 8 tilted Newton iterations, 11.2 seconds end to end on one core. A dense dummy-variable design for the same model would occupy $N \times 9,696$ doubles ≈ 0.4 TB before the first iteration. For verification at feasible scale, the subpopulation of the first 10 municipalities ($n = 97,723$, 196 absorbed groups) was fitted both ways: maximum absolute coefficient deviation 4.0×10^{-11} , tilted 0.1s versus dense 3.0s.

Substantively, the model recovers the calibrated truth within sampling error: income -0.180 (cluster SE 0.005; truth -0.18), pollution $+0.049$ (SE 0.005; truth $+0.06$, within two SEs, identified from within-municipality variation only, as it must be after absorption), female -0.455 (truth -0.45), education gradients $-0.13/ -0.27/ -0.40$ (truth $-0.12/ -0.25/ -0.40$), and a monotone age gradient tracking the Gompertz truth to two decimals. The point of the exercise is not these numbers—the generator made them true—but the demonstration that a model of this shape, with inference, is an interactive-latency computation under the tilted-moment organization and an infeasible one under the dense organization; on genuine register data inside a secure environment, where wall-clock and memory budgets are tight and software dependencies restricted, this difference is what determines whether such models get fitted at all.

7 Discussion

The paper’s message is one computational idea, developed to the point of routine use: when a GLM mixes a rich categorical structure with a few continuous covariates, the exact score and information—hence exact Newton, exact profile Hessians, and exact robust and clustered covariance matrices—are functions of three tilted within-cell moments that one streaming pass produces. Ordinary collapse fails because the likelihood surface needs the whole within-cell distribution (Proposition 1); estimation succeeds because each iteration needs only the current tilt of that distribution (Theorem 1). The experiments quantify what this buys over the implementations in actual use, including the strongest fair competitor (a sparse-design IRLS), and the applications show the method doing routine work on public and register-scale data.

The device composes with, rather than competes against, the fixed-effect absorption stack: absorbed factors handle dummy structure too large even for cells, and the tilted pass handles everything else. The companion theory for that composition—global convergence of the nonlinear alternating-projections cycle for log-concave links, exact local rates for any number of absorbed factors from pairwise weighted cross-tabulations, asymptotics up to the proportional regime $J/N \rightarrow c$ with its Poisson unbiasedness and its logit failure-and-repair, and Firth and penalized variants computed inside the same pass—is developed, proved, and validated in the online supplement; we kept it out of the main text so that the computational contribution stands on its own terms.

Software. The algorithms are implemented twice. The Stata command `stpois` implements the Poisson event-history case in production form (its software article documents the interface); a self-contained Python/NumPy reference implementation (`tilted_glm.py`, roughly 800 lines), which generated every number in this paper and the supplement, accompanies the manuscript and is organized to make the correspondence between code and results line-by-line auditable. Both are available at <https://github.com/andreasljungstrm/stpois>.

Acknowledgments. The author thanks colleagues at SOFI for discussions of register-based event-history workflows that motivated this work.

A Proofs for Section 2

A.1 Proof of Proposition 1

(i) If $\mathbf{x}_i = \bar{\mathbf{x}}_j$ within cells, then $\eta_i = \boldsymbol{\gamma}'\bar{\mathbf{x}}_j + \boldsymbol{\delta}'\mathbf{w}_j + \log t_i$ and (3) becomes $\sum_j \{(\boldsymbol{\gamma}'\bar{\mathbf{x}}_j + \boldsymbol{\delta}'\mathbf{w}_j)D_j - T_j e^{\boldsymbol{\gamma}'\bar{\mathbf{x}}_j + \boldsymbol{\delta}'\mathbf{w}_j}\} + \text{const}$, a function of the displayed cell statistics; the collapsed likelihood is the microdata likelihood up to an additive constant, so their maximizers and Hessians coincide.

(ii) For the Poisson family,

$$\ell(\gamma, \delta) = \sum_j \left\{ \gamma' S_j + \delta' \mathbf{w}_j D_j - e^{\delta' \mathbf{w}_j} \mathcal{L}_{\nu_j}(\gamma) \right\} + \text{const},$$

which establishes computability from $\{S_j, D_j, \mathcal{L}_{\nu_j}\}$. Conversely, suppose a statistic permits evaluation of ℓ on an open set $\mathcal{O} \ni (\gamma_0, \delta_0)$. The functions $\delta \mapsto e^{\delta' \mathbf{w}_j}$ for distinct vectors $\mathbf{w}_j$ are linearly independent on any open set, so from $\delta \mapsto \sum_j e^{\delta' \mathbf{w}_j} \mathcal{L}_{\nu_j}(\gamma)$ one recovers each $\mathcal{L}_{\nu_j}(\gamma)$ for each γ in the open slice. Each ν_j is a finite discrete measure, so $\mathcal{L}_{\nu_j}$ is entire; its values on an open subset of $\mathbb{R}^p$ determine it everywhere by analytic continuation, and the Laplace transform of a finite (positive) measure determines the measure. Hence the statistic determines ν_j , whose minimal description has dimension equal to (one plus p times) its support size.

(iii) Jensen's inequality applied to the strictly convex map $z \mapsto e^z$ under the probability measure ν_j/T_j gives $T_j^{-1} \mathcal{L}_{\nu_j}(\gamma) \geq \exp(\gamma' \bar{\mathbf{x}}_j)$ with equality iff $\gamma' \mathbf{x}$ is ν_j -a.s. constant. $\square$

A.2 Proof of Theorem 1

Differentiating (3) with $\theta_i = \eta_i$: $\phi \partial \ell / \partial \gamma = \sum_i (y_i - b'(\eta_i)) \mathbf{x}_i$ and $\phi \partial \ell / \partial \delta = \sum_i (y_i - b'(\eta_i)) \mathbf{w}_{j(i)}$. Grouping the second sum by cells and using that $\mathbf{w}_j$ is constant within cell j gives $\sum_j \mathbf{w}_j R_j$; grouping the first gives $\sum_i y_i \mathbf{x}_i - \sum_j \mathbf{m}_j^{b'}$ with $\mathbf{m}_j^{b'} = \sum_{i \in j} b'(\eta_i) \mathbf{x}_i = B_j \mathbb{E}_{P_{j, \theta}^{b'}}[\mathbf{x}]$ by the definition of the b' -tilt. Differentiating again, $\phi \mathbf{F} = \sum_i b''(\eta_i) \mathbf{a}_i \mathbf{a}_i'$ with $\mathbf{a}_i = (\mathbf{x}_i', \mathbf{w}_{j(i)}')'$; expanding the blocks and grouping by cells yields (ii), and the moment identities follow from Definition 1. For (iii), $b' = b'' = \exp$ gives $\omega_i = e^{\delta' \mathbf{w}_j} t_i e^{\gamma' \mathbf{x}_i}$, the cell factor cancels in the normalization, and $M_j^{(0)} = e^{\delta' \mathbf{w}_j} \mathcal{L}_{\nu_j}(\gamma)$; the cumulant identities $\mathbb{E}_{P_{j, \gamma}}[\mathbf{x}] = \nabla K_j$ and $\text{Var} = \nabla^2 K_j$ are the standard derivatives of a log-Laplace transform, and $\mathbb{E}[\mathbf{x} \mathbf{x}'] = \text{Var} + \mathbb{E} \mathbb{E}'$ gives the stated γ -block. (iv) The score and information assembled from (i)–(ii) are, entry for entry, the same matrices as those computed from the stacked design; hence so is the Newton step. $\square$

A.3 Proof of Corollary 1

The microdata pass evaluates, per observation, η_i from the cached cell linear predictor plus $\gamma' \mathbf{x}_i$ ($O(p)$), the scalars $b'(\eta_i), b''(\eta_i)$ ($O(1)$), the binned accumulations of ω_i and $\omega_i \mathbf{x}_i$ into cell $j(i)$ ($O(p)$), and the symmetric rank-one update $\omega_i \mathbf{x}_i \mathbf{x}_i'$ ($O(p^2)$ scalar multiply-adds, $p(p+1)/2$ distinct); total $O(N(p^2 + p + 1))$. Assembly of $\sum_j M_j^{(0)} \mathbf{w}_j \mathbf{w}_j'$ and $\sum_j \mathbf{m}_j \mathbf{w}_j'$ costs $O(J(k^2 + kp))$, and the symmetric solve $O((p+k)^3)$. The dense iteration forms $\sum_i \omega_i \mathbf{a}_i \mathbf{a}_i'$ at $O(N(p+k)^2)$ plus the same solve. Ratios follow. With $p = 0$ the pass reduces to two binned sums ($R_j, M_j^{(0)}$), and if the offsets enter through cell sums (pure rate models), even these are fixed after one pass. $\square$

A.4 Proof of Proposition 2

The ML score for a general link is $\sum_i (y_i - \mu_i) \frac{\partial \mu_i / \partial \eta_i}{V(\mu_i)} \mathbf{a}_i = \sum_i (y_i - \mu_i) g'(\mu_i) \omega_i \mathbf{a}_i = \sum_i r_i \mathbf{a}_i$, using $\partial \mu / \partial \eta = 1/g'(\mu)$, and the expected information is $\sum_i \omega_i \mathbf{a}_i \mathbf{a}_i'$ (McCullagh and Nelder, 1989, Sec. 2.5). These have the same cell-decomposable form as in Theorem 1 with (ω_i, r_i) in place of $(b''(\eta_i), y_i - b'(\eta_i))$, so the accumulation and complexity claims carry over. A fixed point of the scoring iteration has $\sum_i r_i \mathbf{a}_i = \mathbf{0}$, i.e., is a root of the ML score equations. For the gamma-log pair, $V(\mu) = \mu^2$ and $g'(\mu) = 1/\mu$ give $\omega_i = 1/(\mu_i^2 \mu_i^{-2}) = 1$; the moments (5) are then constant in $\boldsymbol{\theta}$, and only $R_j = \sum_{i \in j} (y_i - \mu_i) / \mu_i$ requires re-evaluation, an $O(Np)$ pass. $\square$

References

- Bergé, L. (2018). Efficient estimation of maximum likelihood models with multiple fixed-effects: The R package FENmlm. CREA Discussion Paper 2018-13, CREA, University of Luxembourg.
- Clayton, D. and M. Hills (1993). *Statistical Models in Epidemiology*. Oxford: Oxford University Press.
- Correia, S. (2016). A feasible estimator for linear models with multi-way fixed effects. Working paper, Duke University.
- Correia, S., P. Guimarães, and T. Zylkin (2020). Fast Poisson estimation with high-dimensional fixed effects. *The Stata Journal* 20(1), 95–115.
- Daniels, H. E. (1954). Saddlepoint approximations in statistics. *The Annals of Mathematical Statistics* 25(4), 631–650.
- Efron, B. (1981). Nonparametric standard errors and confidence intervals. *Canadian Journal of Statistics* 9(2), 139–158.
- Esscher, F. (1932). On the probability function in the collective theory of risk. *Skandinavisk Aktuarietidskrift* 15, 175–195.
- Gaure, S. (2013). OLS with multiple high dimensional category variables. *Computational Statistics & Data Analysis* 66, 8–18.
- Guimarães, P. and P. Portugal (2010). A simple feasible procedure to fit models with high-dimensional fixed effects. *The Stata Journal* 10(4), 628–649.
- Holford, T. R. (1980). The analysis of rates and of survivorship using log-linear models. *Biometrics* 36(2), 299–305.
- Jensen, J. L. (1995). *Saddlepoint Approximations*. Oxford: Oxford University Press.

- Kitamura, Y. and M. Stutzer (1997). An information-theoretic alternative to generalized method of moments estimation. *Econometrica* 65(4), 861–874.
- Laird, N. and D. Olivier (1981). Covariance analysis of censored survival data using log-linear analysis techniques. *Journal of the American Statistical Association* 76(374), 231–240.
- McCullagh, P. and J. A. Nelder (1989). *Generalized Linear Models* (2nd ed.). London: Chapman & Hall.
- Schennach, S. M. (2007). Point estimation with exponentially tilted empirical likelihood. *The Annals of Statistics* 35(2), 634–672.
- Stammann, A. (2018). Fast and feasible estimation of generalized linear models with high-dimensional k -way fixed effects. *arXiv preprint arXiv:1707.01815*.
- Whitehead, J. (1980). Fitting Cox’s regression model to survival data using GLIM. *Journal of the Royal Statistical Society: Series C* 29(3), 268–275.
- Wickham, H. (2021). *nycflights13: Flights that Departed NYC in 2013*. R package version 1.0.2; data from the U.S. Bureau of Transportation Statistics.